QUADRIGA Data Literacy Framework v4

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The QUADRIGA data literacy framework additionally aims to promote scientific integrity and reflective thinking. It seeks to support an inherently open attitude (in line with the guiding principle of being as open as possible and as closed as necessary) towards the sharing and reuse of research data. To achieve this, the necessary skills are taught at every step of the data-based research process (see data flow).

Prior knowledge of basic concepts of research data management and Open X, as well as knowledge of areas of application for data, are helpful.

data flow	1. Planning			2. Collection and preparation		
competency	**1.1: Identification**	**1.2: Quality assurance**	**1.3: Ethics and legal aspects**	**2.1: Acquisition**	**2.2: Validation**	**2.3: (Pre-)processing**
knowledge	Types of data sources; search and retrieval techniques	Data quality criteria; assessment frameworks	Ethical and legal requirements	Methods and tools for data collection and acquisition	Quality management; data validity	Data formats; conversion methods; data cleaning
skills	Identification of adequate data	Independent evaluation of data quality following defined criteria	Ethical and legally compliant (re-)use of data	Data acquisition; data aggregation	Identification and remediation of data errors and gaps	Data transformation and harmonisation
examples	Query languages, data portals, research data centers, repositories	Data criticism, quality criteria	General Data Protection Regulation (GDPR), good research practice	Collection tools, (import) interfaces	Plausibility, representativeness	Data cleaning tools, mapping procedures

data flow	3. Management		4. Analysis			5. Publication and reuse		
competency	**3.1: Organisation**	**3.2: Metadata**	**4.1: Data analysis**	**4.2: Visualisation**	**4.3: Interpretation**	**5.1: Preservation**	**5.2: Data publication**	**5.3: Communication**
knowledge	Methods, standards, and tools for data and project management	Methods and tools for data description; metadata standards	Methods and tools for data analysis	Methods and tools for data visualisation	Significance and limitations of data analyses and their interdependencies	Requirements and methods for data preservation	Methods and platforms for data publication and sharing for reuse	Methods and strategies of other scholarly publication formats as well as scholarly and science communication
skills	Development of organisational processes, structures, and governance	Selection and application of appropriate standards, description languages, and vocabularies	Selection and application of appropriate analytical methods, processes, and tools	Selection and application of adequate visualisations	Systematic inference of evidence-based conclusions and forecasts	Appraisal of retention value of data; selection of appropriate standards and trustworthy repositories	Ethical, legal, and standard-compliant data publication and sharing	Audience-specific synthesis and communication of research findings
examples	Role allocation, versioning, management systems, data management plan	Annotation, enrichment, (domain-specific) standards, subject classifications, metadata schemas	Analysis tools, reproducibility	Design principles, visualisation tools	Objectivity, transparency	Data and format migration, digital long-term preservation, access control	FAIR principles, persistent identifier systems	Data storytelling, forms of outreach, forms of publication